

Switching Photocatalytic Methane Oxidation Toward Ethanol by Tuning Spin States

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Abstract: The selective oxidation of methane (CH₄) has attractive potentials for mitigating global warming and producing value-added chemicals, whereas the efficient generation of multicarbon products such as ethanol remains challenging, due to the short lifetimes and high unpaired concentrations of reactive intermediates (like •OH, •CH₃, •OCH₃, and •CH₂OH). Herein, we developed a medium-spin Zn–O–Fe(MS) catalyst with tunable Fe(III) spin states, which can efficiently photo-oxidize CH₄ to ethanol at ambient conditions. The unpaired electrons in d orbitals of Fe sites allow for efficient adsorption of •OH, and the *e_g* orbital occupancy enables stabilizing different key carbon-containing intermediates (•OCH₃ versus •CH₂OH). By balancing those two descriptors, the medium-spin Zn–O–Fe(MS) catalyst allows to achieve selective generation of •CH₂OH intermediates with a moderate •OH coverage and subsequent coupling of •CH₃ and •CH₂OH to ethanol. With a light intensity of 100 mW·cm^{−2} and water as a weak oxidant, the Zn–O–Fe(MS) catalyst exhibited a peak photocatalytic CH₄-to-ethanol yield of 372 μmol·g^{−1}·h^{−1} without the use of photosensitizers or sacrificial reagents, which is ~3 times higher than the previously best reported photocatalytic CH₄ oxidation to ethanol under similar conditions. Our study suggests an attractive potential of converting CH₄ to multicarbon products by modulating spin states of the catalytic sites.

Introduction

Methane (CH₄) is the second largest greenhouse gas with continuous global accumulation of >580 million tons annually.^[1] The efficient and controlled conversion of CH₄ to value-added multicarbon (C₂₊) products such as ethanol (CH₃CH₂OH) may not only allow to mitigate the greenhouse effect but also provide useful chemicals for industry and energy storage.^[2] Due to its low polarizability and high stability under ambient conditions,^[3] the CH₄ conversion at present mainly relies on indirect methods to produce syngas by high-temperature reforming, followed by Fischer–Tropsch process to obtain long-chain alkanes,^[4] while this indirect method is intensive in both energy cost and carbon dioxide (CO₂) release, and the product selectivity is also limited.^[5]

The solar-energy driven photocatalysis is a potentially sustainable means of methane oxidation reaction (CH₄OR)

under mild conditions.^[6] Upon the light excitation to photogenerated holes from semiconductor catalysts/electrodes, reactive oxygen species (ROS, such as •OH and •O₂[−]) can be generated from the oxidant to activate the inert C–H bonds of CH₄. However, due to the inefficiency of the C–H bond activation, the concentrations of C₁ intermediates (e.g., •CH₃ and •CH₂OH) are much lower than those ROS concentrations. Thus, the CH₄ conversion is generally dominated by the C₁ product formation pathways, such as CH₃OH,^[7] CH₃OOH,^[8] HCHO,^[9] and HCOOH.^[10] Controlling the ratio between ROS and C₁ intermediates can increase the selectivity of C₂₊ products like ethanol,^[11,12] whereas strong oxidants such as O₂^[11] or hydrogen peroxide (H₂O₂)^[12] are generally needed. Without the addition of a strong oxidant,^[13,14] the ethanol yields have remained low, typically <100 μmol·g^{−1}·h^{−1}.

The activation of C–H bonds in CH₄ involves the change of spin states and is kinetically sluggish.^[15] The spin states of a photocatalyst can affect the electronic orbital occupation of the catalytic sites, which may in turn alter the adsorption of reactive intermediates.^[16] The modulation of spin states of catalytic sites has recently been explored for water splitting,^[17] CO₂ reduction,^[18] and O₂ reduction.^[19] For the CH₄ photocatalysis, theoretical simulations have reported that medium-spin single-atomic Fe(II) catalysts,^[20] high-spin iron-oxygen complexes [FeO]²⁺,^[21] and high-spin single-atomic Ag sites^[15] can have an enhancing effect on the CH₄ activation. For instance, single-atomic Ag sites with a higher spin magnetic moment are beneficial for the formation of spin channels for electron transfer and can thus boost the dissociation of C–H bonds.^[15] Nonetheless, the main selectivities in those theoretical studies are still C₁

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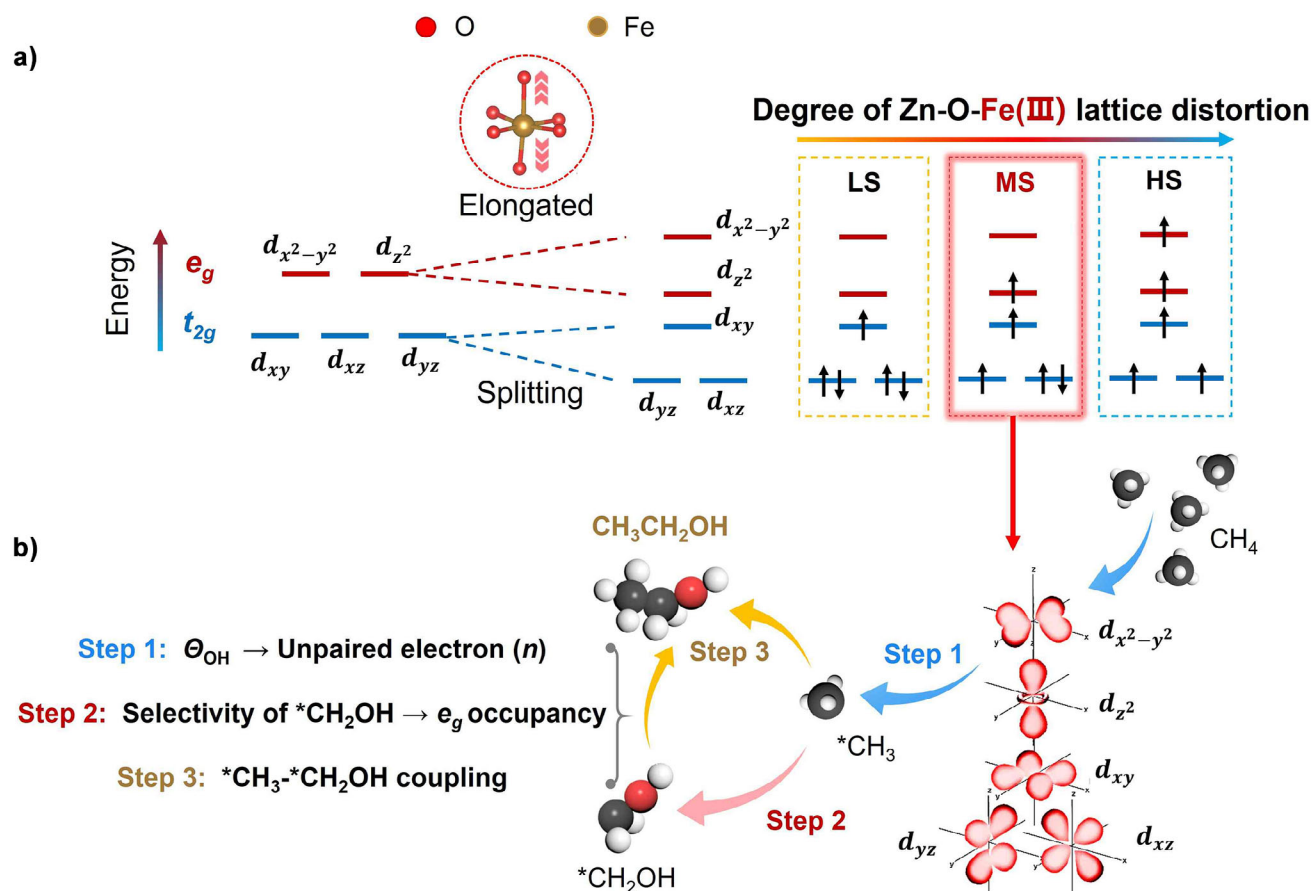


Figure 1. a) Schematic of spin-state tuning of Fe active sites in Zn-Fe complex oxides by modulation of the Zn-O-Fe(III) lattice distortion degree. b) Schematic of the photocatalytic CH_4 oxidation to $\text{CH}_3\text{CH}_2\text{OH}$ product on the Zn-O-Fe(MS) photocatalyst with medium-spin Fe sites via balancing the descriptors of unpaired electrons (n) and e_g occupancy.

products such as methanol. Thus, although adjusting the spin states of catalytic sites may trigger a reactive environment to activate CH_4 without the need for high-concentration ROS, the capability of selective CH_4 -to- C_2+ conversion by spin tuning has not been realized.

In this work, we demonstrate a spin-state tuning strategy on Zn-Fe complex oxides with the Zn-O-Fe lattice distortion, which modulates the unpaired electron numbers (n) and the orbital occupancy in the e_g orbitals of Fe sites (Figure 1a). These two structural features enable to tune not only the $^*\text{OH}$ coverage (θ_{OH}) at the catalytic interface but also the activation of CH_4 toward different C_1 intermediates, thus leading to different reaction pathways. Notably, the medium-spin Zn-Fe complex oxide (designated as Zn-O-Fe(MS)) allows to activate CH_4 to $^*\text{CH}_2\text{OH}$, followed by the C-C coupling and $^*\text{OH}$ attack to produce ethanol (Figure 1b). In contrast, the low-spin (Zn-O-Fe(LS)) and high-spin (Zn-O-Fe(HS)) catalysts show distinctive selectivities of C_1 products as CH_3OH and CH_3OOH , respectively. Under $100 \text{ mW}\cdot\text{cm}^{-2}$ light intensity and only using water as a weak oxidant, the Zn-O-Fe(MS) photocatalyst exhibited a peak CH_4 -to-ethanol yield of $372 \mu\text{mol}\cdot\text{g}^{-1}\cdot\text{h}^{-1}$ without photosensitizers or sacrificial reagents, corresponding to ~ 3 times of increase compared to the best CH_4OR reported previously under

similar conditions, thus featuring attractive potentials of modulating spin states to promote the photocatalytic CH_4 -to- C_2+ conversion.

Results and Discussion

The three Zn-Fe complex oxides with different Fe spin states were synthesized by the combination of hydrothermal and calcination methods (Methods in the Supporting Information), designated as Zn-O-Fe(LS), Zn-O-Fe(MS), and Zn-O-Fe(HS), respectively. The molar ratios of Fe and Zn in all those three samples were controlled at 1:10, confirmed by inductively coupled plasma (ICP). The X-ray diffraction (XRD) patterns of all samples presented the ZnO hexagonal wurtzite phase (JCPDS No. 36-1451, Figure 2a). The crystal structures of Fe were attributed to Fe_2O_3 (JCPDS No. 33-0664), ZnFe_2O_4 (JCPDS No. 22-1012), and Fe_3O_4 (JCPDS No. 19-0629) for Zn-O-Fe(LS), Zn-O-Fe(MS), and Zn-O-Fe(HS), respectively (Figure 2a), in accord with their transmission electron microscopy (TEM) images (Figures 2b and S1). Energy-dispersive spectroscopy (EDS) exhibited the distributions of Zn, Fe, and O elements of all those samples (Figures S2-S4). In addition, the Raman spectra of ZnO

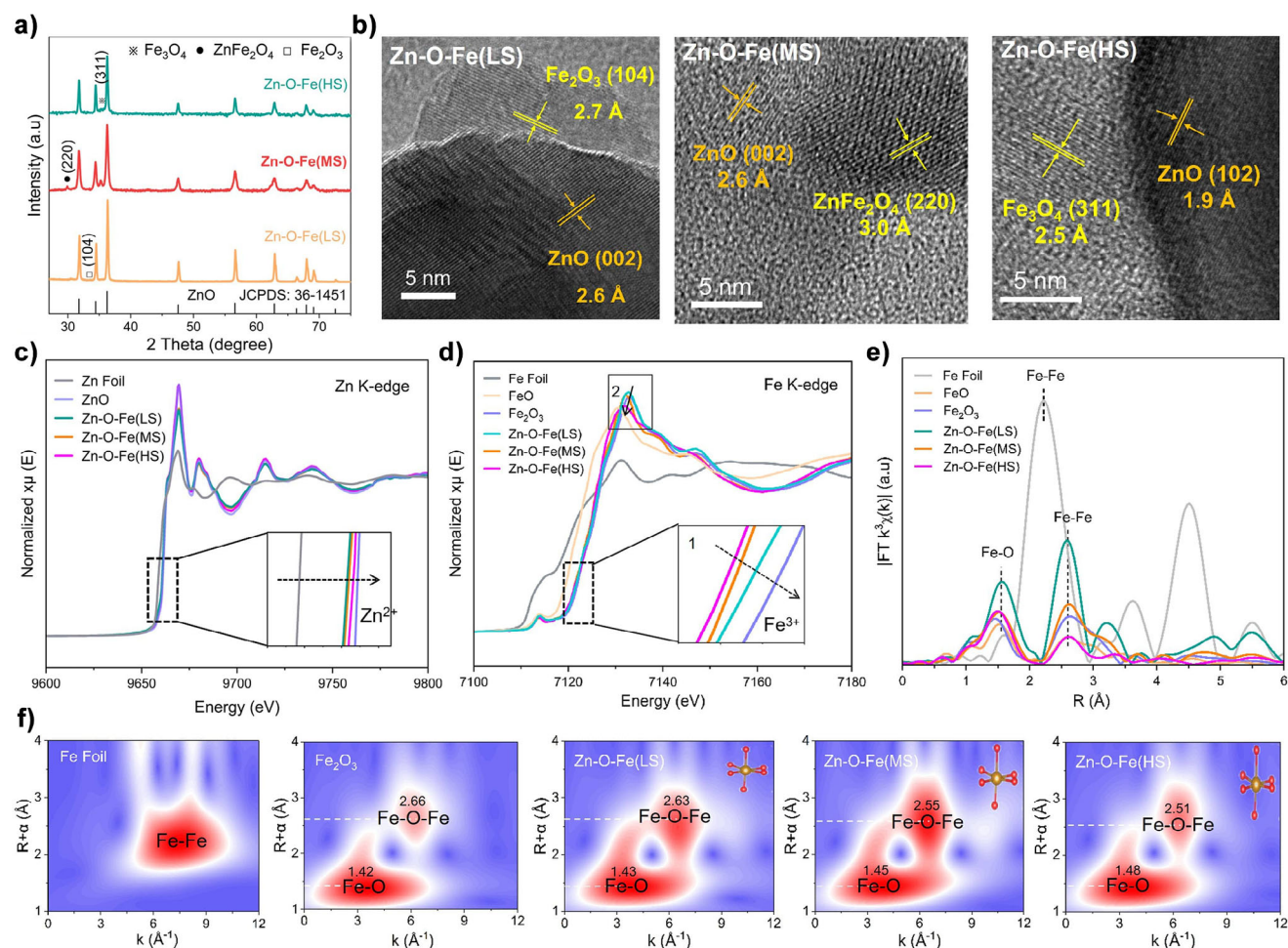


Figure 2. a) XRD patterns and b) HRTEM images of Zn–O–Fe(LS), Zn–O–Fe(MS), and Zn–O–Fe(HS) photocatalysts. Normalized c) Zn K-edge and d) Fe K-edge XANES spectra of Zn–O–Fe(LS), Zn–O–Fe(MS), Zn–O–Fe(HS), as well as Fe, FeO, and Fe₂O₃ standard samples. e) Fourier-transform k^3 -weighted Fe K-edge EXAFS spectra. f) Normalized Fe K-edge XANES spectra Wavelet transforms.

and all those Zn–O–Fe oxide samples displayed the major peaks of ZnO at 333, 437, and 576 cm^{-1} (Figure S5). The peaks in the range of 650–670 cm^{-1} were attributed to the Fe–O vibration of Fe₂O₃ in Zn–O–Fe(LS), Zn–O vibration of ZnFe₂O₄ in Zn–O–Fe(MS), and Fe–O vibration of Fe₃O₄ in Zn–O–Fe(HS), respectively.

X-ray absorption near-edge structure (XANES) analysis was performed to analyze the valence states and coordination environments of Fe and Zn. The absorption edge of Zn K-edge of those three samples were close to that of ZnO (Figure 2c), indicating the oxidation states as ~ 2 . The valence of Zn in the Zn–O–Fe(LS), Zn–O–Fe(MS), and Zn–O–Fe(HS) catalysts increased sequentially with the change of iron oxides, reflecting the enhanced interaction between Zn–Fe sites in the composite catalysts (Figure S6). The normalized Fe K-edge XANES spectra showed that the Fe valence states of those three catalysts were between FeO and Fe₂O₃ (Figure 2d), and the shift of the Fe absorption edge to higher energies was correlated to the valence state,^[22] in the order of: Zn–O–Fe(HS) < Zn–O–Fe(MS) < Zn–O–Fe(LS) < Fe₂O₃ (inset-1 in Figure 2d). In addition, the intensity reduction of white line peak, associated with the electron transition from

1s to 4p,^[23] followed the order of: Zn–O–Fe(LS) < Zn–O–Fe(MS) < Zn–O–Fe(HS) (inset-2 in Figure 2d), suggesting the increased degree of elongated orbital distortion due to the axial lattice stretching.^[24] The Fourier transform extended X-ray absorption fine structure (EXAFS) spectra of the three Zn–O–Fe samples showed two main peaks at ~ 1.5 and ~ 2.6 Å (Figure 2e), assigned to the Fe–O^[25] and Fe–Fe bonds,^[26] respectively. Wavelet-transform (WT) contour plots of the Fe K-edge EXAFS exhibited the elongation of Fe–O bonds and shortening of Fe–Fe bonds in the order of: Zn–O–Fe(LS) $\rightarrow$ Zn–O–Fe(MS) $\rightarrow$ Zn–O–Fe(HS) (Figure 2f and Table S1), confirming that the Fe–O octahedral structure was axially stretched and laterally compressed (inset in Figure 2f), similar to previous reports.^[23]

The X-ray photoelectron spectroscopy (XPS) of the Fe 3s core-level signal was deconvoluted into two subpeaks centered ~ 91 and 93 eV (Figure 3a), attributed to the coexistence of two spin states of Fe atoms.^[27] The proportion of unfilled 3d orbitals presented a decreasing trend from Zn–O–Fe(LS) as 36% to Zn–O–Fe(HS) as 19%, suggesting the gradual increase of the 3d orbital occupancy of Fe sites.^[28] The electronic structures of unpaired electrons were further

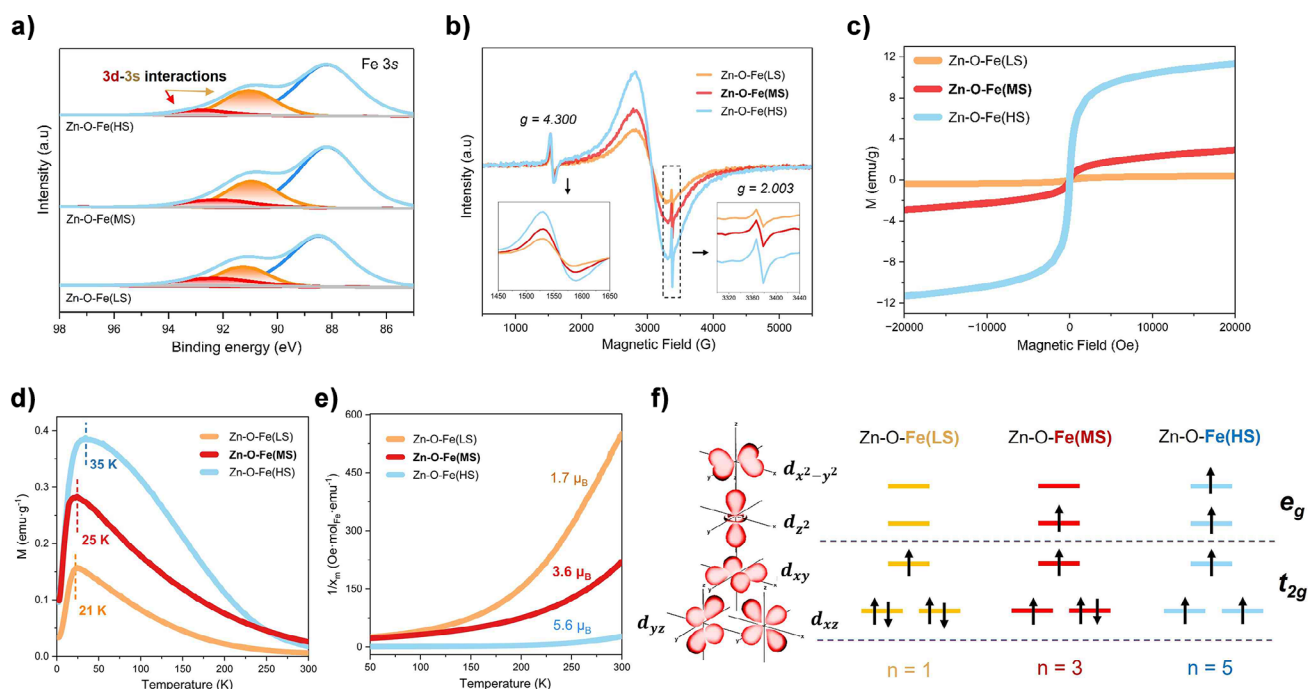


Figure 3. a) Fe 3s XPS spectra. b) EPR spectra measured at 77 K. c) Magnetization (M - H) curves measured at 300 K. Temperature-dependent magnetic susceptibility d) M - T and e) $1/\chi_m$ - T curves, and f) schematic of electronic configurations of Zn-O-Fe(LS), Zn-O-Fe(MS), and Zn-O-Fe(HS) photocatalysts.

characterized by electron paramagnetic resonance (EPR) spectroscopy (Figure 3b). The signal intensities at g -factors of 4.3 and 2.0 represented the densities of unpaired electrons in the tetrahedral and octahedral structures occupied by Fe^{3+} , respectively,^[29] and the broad peak at $g = 2.2$ was attributed to the radiatively disrupted centers caused by magnetic interference signals.^[30] The spin degrees of both Fe sites showed an increasing order as: Zn-O-Fe(LS) < Zn-O-Fe(MS) < Zn-O-Fe(HS), in agreement with the XPS results.

The unpaired electron numbers of those three samples were further identified by measuring the magnetic properties with superconducting quantum interference device (SQUID, Figure 3c). The saturation magnetization (M_s , the intercept of the magnetic hysteresis loop on the y axis) of those samples gradually increased in the order as: Zn-O-Fe(LS) $\rightarrow$ Zn-O-Fe(MS) $\rightarrow$ Zn-O-Fe(HS). Further analysis of the positive and negative coercivity fields (i.e., the intercept of the magnetic hysteresis loop on the x axis) revealed the enhancement of their exchange bias fields (HEB) from 0.4 to 3.8 Oe (Figure S7), indicating the increase of the unpaired electron number of the Fe sites.^[31] From the magnetization versus temperature plots measured under zero-field-cooling conditions with an applied magnetic field of 300 Oe, all three Zn-Fe oxide samples exhibited an increasing and then decreasing trend of the magnetization strength with temperature (Figure 3d). The Néel temperature (i.e., the peak of the curve)^[32] increased from 21 to 35 K in the order as: Zn-O-Fe(LS) $\rightarrow$ Zn-O-Fe(MS) $\rightarrow$ Zn-O-Fe(HS), confirming the enhanced exchange interaction within the samples.^[33] By fitting the $1/\chi_m$ versus

T curve (Figure 3e), the unpaired electron numbers can be obtained according to the following equation: $2.828 \sqrt{\chi_m T} = \mu_{\text{eff}} = \sqrt{n(n+2)}$,^[34] where χ_m is the magnetic susceptibility ($\text{emu mol}_{\text{Fe}}^{-1} \text{Oe}^{-1}$), μ_{eff} is effective magnetic moment (μ_B), n is unpaired d electrons, and T is absolute temperature (K). The effective magnetic moments of Fe in the Zn-O-Fe(LS), Zn-O-Fe(MS), and Zn-O-Fe(HS) samples were measured as 1.7, 3.6, and $5.6 \mu_B$, respectively, and thus the unpaired d electron number (n) were calculated as (Figure 3f): 1 for Zn-O-Fe(LS) (i.e., $t_{2g}^5 e_g^0$, low-spin), 3 for Zn-O-Fe(MS) (i.e., $t_{2g}^4 e_g^1$, medium-spin), and 5 for Zn-O-Fe(HS) (i.e., $t_{2g}^3 e_g^2$, high-spin). The different spin numbers and orbital occupancy were ascribed to the axial stretching and orbital splitting of Fe in those samples.

The photocatalytic CH_4OR was then conducted at room temperature using H_2O as a weak oxidant (Methods in the Supporting Information), and no photosensitizers or sacrificial reagents were used in this work. The bandgaps of those three samples were first calculated by UV-visible spectroscopy (Figure S8a,b), and then their valence band edges were measured by valence band-XPS (Figure S8c), showing that all three samples had energy band structures capable of activating both H_2O and CH_4 (Figure S8d). The gas and liquid products were quantified by gas chromatography and ^1H -nuclear magnetic resonance, respectively, and included CH_3OH , CH_3OOH , and $\text{CH}_3\text{CH}_2\text{OH}$ (Figure 4a). The main liquid product selectivity changed from CH_3OH (76%) for Zn-O-Fe(LS), to $\text{CH}_3\text{CH}_2\text{OH}$ (67%) for Zn-O-Fe(MS), and then to CH_3OOH (79%) for Zn-O-Fe(HS) (Figure S9). The Zn-O-Fe(HS) catalyst presented the highest

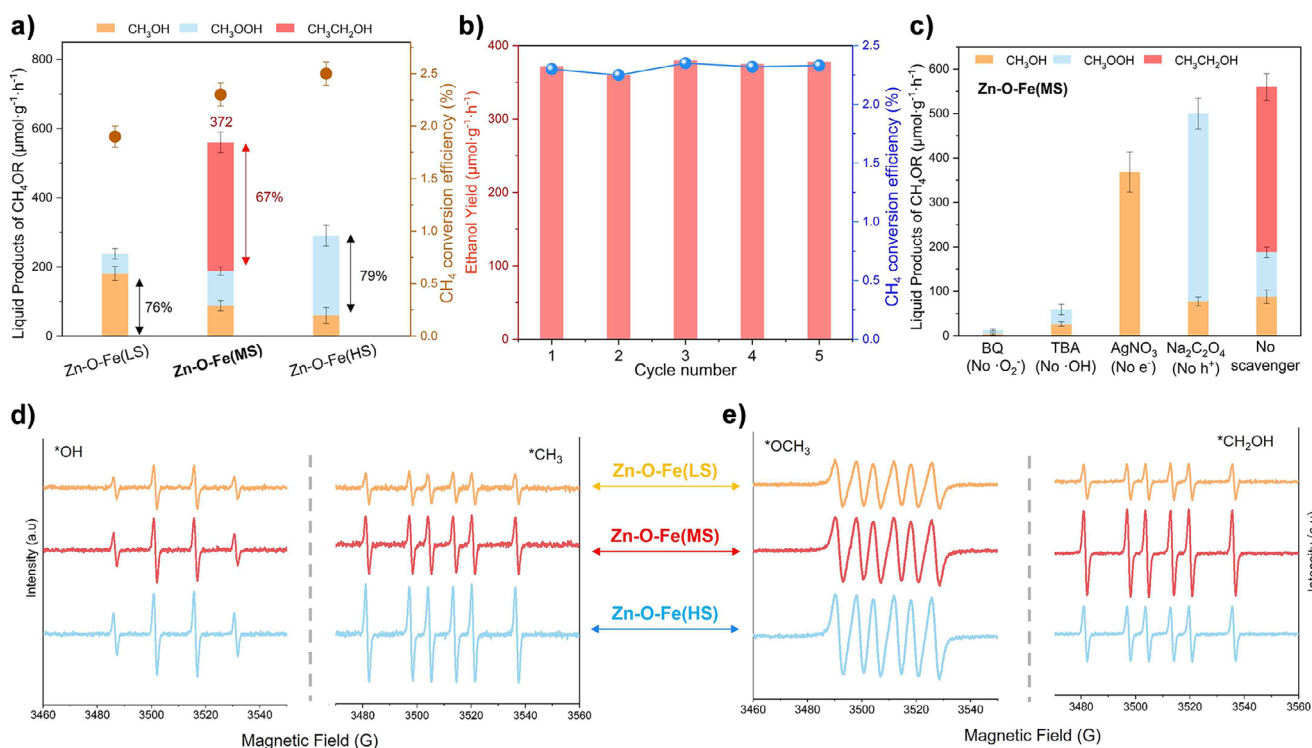


Figure 4. a) Left y-axis: photocatalytic production rates (color bars) of liquid products on Zn–O–Fe(LS), Zn–O–Fe(MS), and Zn–O–Fe(HS) photocatalysts. Right y-axis: conversion efficiency of CH_4 (brown dots) on Zn–O–Fe(LS), Zn–O–Fe(MS), and Zn–O–Fe(HS) photocatalysts. b) Cyclic photocatalytic CH_4OR stability on Zn–O–Fe(MS). c) Production rates of liquid products on Zn–O–Fe(MS) with the separate addition of benzoquinone (BQ), *tert*-butylalcohol (TBA), silver nitrate (AgNO_3), or sodium oxalate ($\text{Na}_2\text{C}_2\text{O}_4$). Error bars represent standard deviations obtained from three independent measurements. d) EPR signals of $\text{DMPO}\cdot\text{OH}$ and $\text{DMPO}\cdot\text{CH}_3$ adducts under photocatalytic CH_4OR condition. e) EPR signals of $\text{DMPO}\cdot\text{OCH}_3$ and $\text{DMPO}\cdot\text{CH}_2\text{OH}$ under photocatalytic CH_3OH oxidation condition.

CH_4 conversion rate (Figure 4a, right y axis), in good accord with its highest photocurrent density (Figure S10a) and lowest electrochemical impedance (Figure S10b). On the other hand, the Zn–O–Fe(MS) catalyst exhibited the highest $\text{CH}_3\text{CH}_2\text{OH}$ production rate of 372 $\mu\text{mol}\cdot\text{g}^{-1}\cdot\text{h}^{-1}$, corresponding to ~ 3 times of the best reported data using water as weak oxidant [12,35] (Table S2). In comparison, for pure ZnO or ZnFe_2O_4 (Figure S11a,b), the CH_4OR yields of both C_1 and C_2 products decreased dramatically, especially for ethanol (Figure S11c).

The effect of the composite ratio between ZnFe_2O_4 and ZnO on the CH_4OR selectivity was then investigated. When the $\text{ZnFe}_2\text{O}_4/\text{ZnO}$ ratio was tuned from 5% to 15%, the spin state of Fe sites changed from low-spin to medium-spin and then high-spin (Figure S12a,b). The major CH_4OR products of those photocatalysts presented a similar trend (Figure S12c), in which the highest ethanol selectivity was still produced from the $\text{ZnFe}_2\text{O}_4/\text{ZnO}$ composite with medium-spin, confirming the effect of spin state on the product selectivity.

Furthermore, the photocatalytic CH_4OR stability of the Zn–O–Fe(MS) catalyst showed that the ethanol production rate and CH_4 conversion ratio were maintained for >15 h (i.e., each cycle of 3 h, for 5 consecutive cycles, Figure 4b). After the photocatalysis, the XRD, TEM, EDS mapping, and STEM results revealed that the Zn–O–Fe(MS) catalyst still retained

its original morphology, composition, and structure (Figures S13 and S14). The Fe 2p XPS peaks shifted slightly to higher binding energies, and the Zn 2p peaks shifted slightly to lower binding energies (Figure S15), suggesting the migration of photogenerated holes and electrons to the Fe and Zn sites, respectively.

To identify the key radicals involved in the photocatalytic CH_4OR , 1,4-benzoquinone (BQ), *tert*-butylalcohol (TBA), silver nitrate (AgNO_3), and sodium oxalate ($\text{Na}_2\text{C}_2\text{O}_4$) were separately added into the photocatalytic reaction as scavengers (Figure 4c), which selectively eliminated superoxide radical ($\cdot\text{O}_2^-$), hydroxyl radicals ($\cdot\text{OH}$), photogenerated electrons (e^-), and photogenerated holes (h^+), respectively.[36] No $\text{CH}_3\text{CH}_2\text{OH}$ product was detected after the addition of any scavenger, suggesting that the photocatalytic CH_4 -to-ethanol in this work was mediated by free radicals. To further investigate the mechanism of product selectivity, the as-formed active intermediates during the photocatalysis were quantified by EPR spectroscopy, using 5,5-dimethyl-1-pyrroline *N*-oxide (DMPO) to capture $\cdot\text{OH}$, $\cdot\text{CH}_3$, $\cdot\text{OCH}_3$, and $\cdot\text{CH}_2\text{OH}$ (Methods in the Supporting Information). With the spin degree increase of the Fe sites, the $\cdot\text{CH}_3$ signal intensity was gradually enhanced and synchronized with the $\cdot\text{OH}$ signal intensity (Figure 4d), confirming the increase of $\cdot\text{OH}$ coverage facilitated the CH_4 conversion.

Considering that CH_3OH may serve as a key intermediate to form $\text{CH}_3\text{CH}_2\text{OH}$ [37] and CH_3OOH ,[38] the photocatalytic

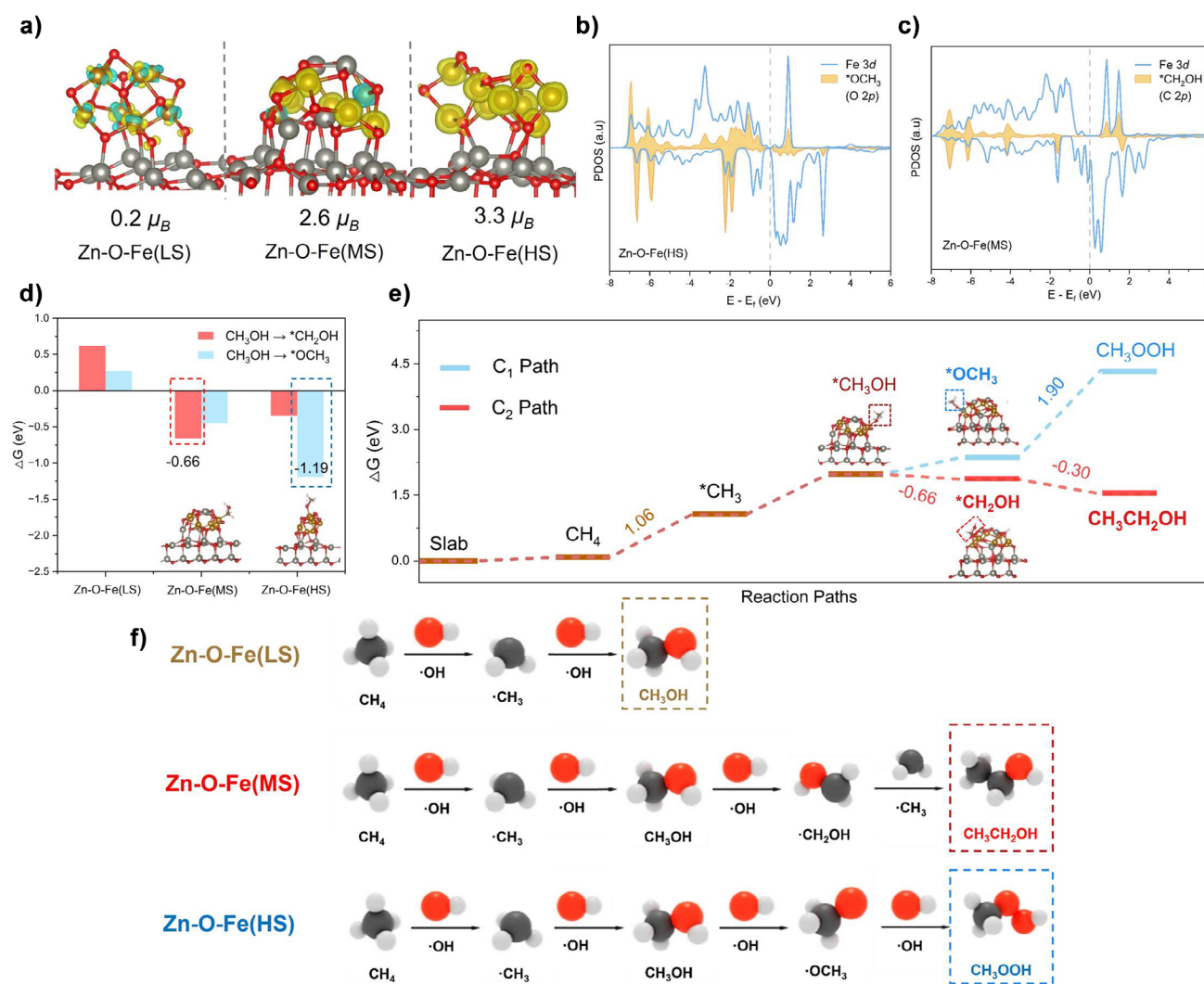


Figure 5. a) Schematic illustration of the effective magnetic moments and spin densities of the Fe sites on Zn–O–Fe(LS), Zn–O–Fe(MS), and Zn–O–Fe(HS). Blue and yellow shadings represent the electron spins. b) PDOS of Fe 3d and O 2p in *OCH_3 on Zn–O–Fe(HS). c) PDOS of Fe 3d and C 2p in $\text{*CH}_2\text{OH}$ in Zn–O–Fe(MS). d) Activation energies of CH_3OH to $\text{*CH}_2\text{OH}$ or to *OCH_3 on Zn–O–Fe(LS), Zn–O–Fe(MS), and Zn–O–Fe(HS), respectively. e) Two complete reaction pathways for the generation of C_1 or C_2 products in Zn–O–Fe(MS). f) Reaction pathways for the photocatalytic conversion of CH_4 on Zn–O–Fe(LS), Zn–O–Fe(MS), and Zn–O–Fe(HS). The white, red, and black spheres represent H, O, and C atoms.

CH_3OH oxidation was explored by detecting the EPR signals of *OCH_3 and $\text{*CH}_2\text{OH}$ by those three samples (Figure 4e). For Zn–O–Fe(LS), both the *OCH_3 and $\text{*CH}_2\text{OH}$ signals were the lowest among those three samples, in accord with its main CH_4OR product as CH_3OH . For Zn–O–Fe(HS), it presented the highest *OCH_3 signal that was originated from the decomposition of CH_3OH , which was further converted into CH_3OOH as the main product.^[39] In contrast, the Zn–O–Fe(MS) catalyst presented the highest $\text{*CH}_2\text{OH}$ signal, which was subsequently converted into $\text{CH}_3\text{CH}_2\text{OH}$ as the main product.^[40] Furthermore, the increase of water amount resulted in decreasing the $\text{CH}_3\text{CH}_2\text{OH}$ product and increasing the CH_3OOH product (Figure S16), suggesting the contribution of the *OH concentration to form $\text{CH}_3\text{CH}_2\text{OH}$. The effect of *OH concentration on the product selectivity of the Zn–O–Fe(MS) catalyst was further compared by adding O_2 as a strong oxidant with water (Figure S17).

Both the CH_4 conversion rate and the CH_3OOH yield increased, whereas the ethanol yield decreased, suggesting that the ethanol production required both medium-spin Fe sites and a moderate *OH concentration, whereas a higher *OH concentration favored the CH_3OOH production.

Density functional theory (DFT) calculations were further conducted to investigate the CH_4OR mechanism. To mimic the same unpaired electron numbers with different catalysts (i.e., $n = 1, 3$, and 5), the composites of ZnO with Fe_2O_3 , ZnFe_2O_4 , and Fe_3O_4 were constructed to represent Zn–O–Fe(LS), Zn–O–Fe(MS), and Zn–O–Fe(HS), respectively (Figure S18). The spin electron density of the Fe sites in the catalyst is increased sequentially, and the magnetic moment increases from 0.2 to 3.3 μ_B (Figure 5a). The projected density of states (PDOS) was first calculated to investigate the spin selectivity of $\text{*CH}_2\text{OH}$ or *OCH_3 intermediate for different catalysts. The overlap between the 3d orbital of Fe site and

the p orbitals of $^*\text{OCH}_3$ (Figures 5b and S19) on the high-spin Zn–O–Fe(HS) is much stronger than that on the medium-spin Zn–O–Fe(MS) (Figures S20 and S21), suggesting the preferential stabilization of the $^*\text{OCH}_3$ intermediate by Zn–O–Fe(HS). On the contrary, the d orbitals of Fe sites present a stronger overlap with the p orbitals of $^*\text{CH}_2\text{OH}$ (Figures 5c and S22) on the medium-spin Zn–O–Fe(MS) than that on the high-spin Zn–O–Fe(HS) (Figures S23 and S24), in accord with the experimental observation of the formation of $^*\text{CH}_2\text{OH}$ on the Zn–O–Fe(MS) catalyst. Thus, the different spin number and d orbital occupancy can affect the selective generation of different intermediates (i.e., $^*\text{OCH}_3$ versus $^*\text{CH}_2\text{OH}$) during photocatalysis.

The free energy changes of the three key steps of the CH_4 -to-ethanol conversion were then investigated, including CH_4 -to- $^*\text{CH}_3$,^[11] $^*\text{CH}_3\text{OH}$ -to- $^*\text{CH}_2\text{OH}$, or $^*\text{CH}_3\text{OH}$ -to- $^*\text{OCH}_3$,^[41] and the coupling of $^*\text{CH}_3$ and $^*\text{CH}_2\text{OH}$ to form ethanol.^[12] (Here * stands for the adsorbed intermediate.) First, for the CH_4 -to- $^*\text{CH}_3$ step, the free energy change on Zn–O–Fe(LS), Zn–O–Fe(MS), and Zn–O–Fe(HS) were calculated as 1.26, 1.06, and 0.89 eV, respectively (Figure S25), suggesting that as the spin degree of Fe sites increases, the CH_4 activation to $^*\text{CH}_3$ is more favorable. Next, the $^*\text{CH}_3\text{OH}$ -to- $^*\text{CH}_2\text{OH}$ or $^*\text{OCH}_3$ is the key step for the conversion of ethanol or CH_3OOH , respectively.^[8,42] The lowest free energy change on Zn–O–Fe(MS) was found as -0.66 eV for $^*\text{CH}_3\text{OH}$ -to- $^*\text{CH}_2\text{OH}$ (Figures 5d and S26) and on Zn–O–Fe(HS) as -1.19 eV for $^*\text{CH}_3\text{OH}$ -to- $^*\text{OCH}_3$ (Figures 5d and S27), indicating that the spin-state of the Fe sites can lead to different CH_4 oxidation pathways. Third, for the $^*\text{CH}_3$ - $^*\text{CH}_2\text{OH}$ coupling to ethanol pathway (Figures S28 and S29), the Zn–O–Fe(MS) catalyst presents the lowest free energy change of -0.3 eV compared to that of Zn–O–Fe(LS) and Zn–O–Fe(HS), also confirming the most favorable ethanol selectivity on the medium-spin Fe sites.

To further investigate whether the formation of $^*\text{CH}_2\text{OH}$ or $^*\text{OCH}_3$ is followed by the C–C coupling or further oxidation by $^*\text{OH}$, the full CH_4 conversion pathways were calculated on Zn–O–Fe(MS) (Figure 5e). The reaction site for water activation to produce $^*\text{OH}$ ($\text{H}_2\text{O} \rightarrow ^*\text{OH} + ^*\text{H}$) on Zn–O–Fe(MS) was first calculated (Figure S30), revealing that Zn sites at the composite interface have the lowest free energy change (-0.45 eV, Figure S31). The CH_4 -to- $^*\text{CH}_3$ step, as the activation step, has a free energy change of 1.06 eV. Afterward, if the $^*\text{OCH}_3$ intermediate is formed, either subsequent coupling with $^*\text{CH}_3$ (2.70 eV) or reaction with $^*\text{OH}$ (1.90 eV) is a highly endothermic step (Figures S32 and S33). In contrast, when the $^*\text{CH}_2\text{OH}$ intermediate is formed, the subsequent coupling with $^*\text{CH}_3$ presents a free energy change of -0.30 eV (Figures S34 and S35), confirming the preferable ethanol formation pathway on Zn–O–Fe(MS).

Based on the aforementioned experimental and computational results, the mechanism of photocatalytic CH_4OR catalyzed by different spin-state Fe sites is proposed (Figure 5f and Table S3). The configuration of unpaired electrons (i.e., both the unpaired electron number n and e_g orbital occupancy) can influence the selective adsorption of $^*\text{C}_1$ intermediates, and alter the formation pathways of different

products. For the low-spin Zn–O–Fe(LS) catalyst (where $n = 1$, e_g occupancy = 0), both the surface $^*\text{OH}$ coverage (θ_{OH}) and $^*\text{CH}_3$ are low, and thus the main product selectivity is methanol. For the medium-spin Zn–O–Fe(MS) catalyst (where $n = 3$, e_g occupancy = 1) with a medium θ_{OH} , the d orbitals of Fe sites overlap more efficiently with the p orbitals of the $^*\text{CH}_2\text{OH}$ intermediate, and thus ethanol can be produced by effective coupling of both $^*\text{CH}_2\text{OH}$ and $^*\text{CH}_3$. For the high-spin Zn–O–Fe(HS) catalyst (where $n = 5$, e_g occupancy = 2) with a high θ_{OH} , the d orbital of Fe sites can overlap with the $^*\text{OCH}_3$ intermediate more efficiently, and thus the fast coupling of $^*\text{OCH}_3$ and $^*\text{OH}$ results in the formation of CH_3OOH .

Conclusions

In summary, we have developed a Zn–O–Fe(MS) photocatalyst with both medium-spin Fe sites and medium e_g orbital occupancy, which allow to efficiently activate CH_4 to $^*\text{CH}_3$ and then the $^*\text{CH}_2\text{OH}$ intermediate, followed by the coupling of $^*\text{CH}_2\text{OH}$ and $^*\text{CH}_3$ to produce ethanol. In contrast, Zn–O–Fe photocatalysts with low-spin or high-spin show substantially different selectivities of C_1 products as CH_3OH and CH_3OOH , respectively. Without the additional strong oxidants, photosensitizers, or sacrificial reagents, the Zn–O–Fe(MS) photocatalyst exhibited a high CH_4 -to-ethanol conversion rate of $372 \mu\text{mol}\cdot\text{g}^{-1}\cdot\text{h}^{-1}$, substantially exceeding the best CH_4OR reported previously under similar conditions. Our study suggests attractive new potentials of tuning spin states for converting CH_4 into multicarbon products.

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Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords: C–C coupling • CH_4 oxidation • Intermediates • Photocatalysis • Spin state

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